Presenter Minyan Xu

Sampling 50 Author Lists for Every Paper

DONE All 9,108 papers · 455,400 author lists · Qwen3-32B, bf16, thinking off

1 Ask
Qwen3-32B gets the title and year; no search, no retrieval

2 Sample
50 lists per paper at T = 1, with no top-p or top-k cut

3 Match
The paper's own matcher: last name + first initial

4 Pool
All names from the 50 lists, with counts

Our default answer reproduces the paper

	Ours	Paper's run
True authors named (9,108 papers)	**1,211**	1,202
Papers mostly named (HR2 ≤ 0.5)	**222**	222
Mean HR2	**0.9691**	0.9691

Two chance baselines

- Common names match by luck: 50 lists often include a "J. Wang"
- Random: authors of 3 other papers, same field and citation bin
- Topic (stricter): authors of the 3 most similar titles in the field
- Only matches above the baseline count

Same model, prompt and matcher as the paper's Qwen3-32B run; every count is net of a chance baseline.

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The Recognition Test Leaks; Own Wrong Lists Are Harder

Papers whose true list is chosen in ≥ 3 of 4 option orders (%)



Floor ≈ 16%, not 5%

With another paper's title, 16% still pass: the model prefers some lists in every order.

Hidden title: a year cue

Teams from the same years cut it from 24% to 18%, close to the floor.

Own wrong lists: much harder

Known falls from 70% to 41%: 2,009 of 6,179 known papers turn unknown; 82 move up.

Labels rebuilt with the team's rules on our runs: recalled 222 · known 6,179 · notsure 924 · unknown 1,783. Controls: 1,023 papers.

Read "known" against the 16% swapped-title floor; with the model's own wrong lists, a third of known papers turn unknown.

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Known Papers: The Right Author Is Often in the Samples

Known · 6,179 papers

Named by default **11.4**

+64.5 more true authors per 100 papers in the 50 samples, net of the topic baseline

[61.8, 67.6] · 2,625 papers (42%)

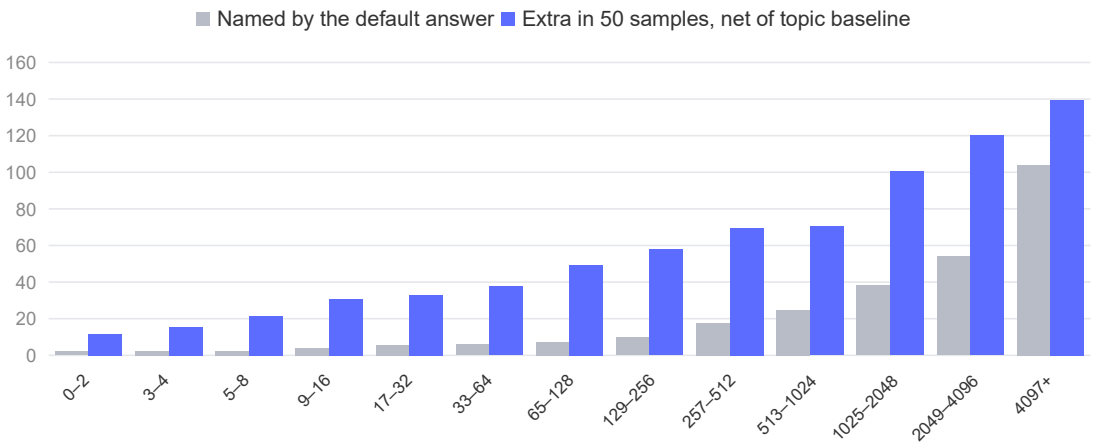
Unknown · 1,783 papers

Named by default **1.3**

+12.9 more true authors per 100 papers in the 50 samples, net of the topic baseline

[10.1, 15.9] · 254 papers (14%)

True authors per 100 papers, by citation bin (all 9,108 papers)



More samples reach more (1,023-paper subset): 50 → 100 → 200 samples name 24% → 30% → 36% of true authors; topic baseline 2% → 3% → 4%.

At 50 samples, 42% of known papers have a true author the default answer misses but some sample names.

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Counting Beats Asking: Frequency Is the Selector to Beat

1 **Count: how many of the 50 samples name it** — AUC 0.74

Ranks a true candidate above a false one of the same paper 74% of the time; top candidate true for 43%.

2 **Ask: “Is X an author?” (P(True))** — AUC 0.61

Worse than counting: the model’s own yes/no adds nothing beyond frequency; top candidate true for 31%.

3 **Choose among the top 4 + “none of these”** — +3.0 pts

Picks a true author 58.9% vs 55.9% for the most frequent [+0.2, +5.8]; says “none” only 30% of the time when none is true.

Test split · 6,374 papers

Default answer vs names in ≥ 7 of 50 samples (rule chosen on the tune split)

True authors / 100 papers

13.4 → 17.2 **+29%**

Invented names / 100 papers

516 → 101 **−80%**

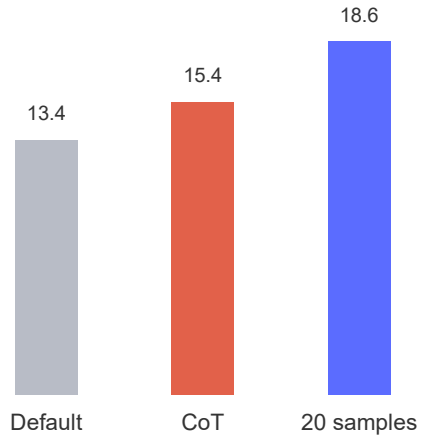
Names that recur across samples are usually true: more true authors and far fewer invented ones than the default answer.

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At Equal Compute, Counting Matches Thinking

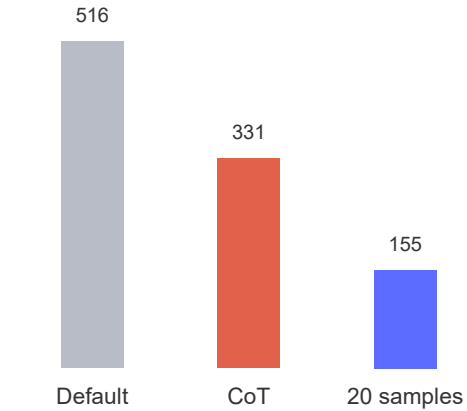
True authors / 100 papers

tokens per paper: 33 · 646 · 645



Invented names / 100 papers

tokens per paper: 33 · 646 · 645



Test split, 6,374 papers. Per token, ordinary samples reach 2.5× more true authors than thinking answers.

Paired: 20 samples + count minus CoT

Mean HR2 +0.002 [−0.001, +0.005], n.s.

True authors +3.2 per 100 [+1.9, +4.4]

Invented names −176 per 100 [−233, −131]

At ~3,200 tokens (1,023-paper subset)

5 CoT answers + vote: best HR2 (−0.016) and fewest invented names. 100 samples + count: more true authors (+5.4 per 100).

At thinking mode's cost, counting over 20 samples ties it on HR2, names more true authors and invents half as many.

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In the Steering Unit: What Selection Adds on Known Papers

Known test papers (4,271; our labels): true authors named per 100 papers

Method	True / 100	vs default [95% CI]	Invented / 100
Default answer	11.3	—	529
Thinking (CoT), 1 answer	16.5	+5.2 [+3.7, +6.6]	345
Names in ≥ 3 of 20 samples	**19.3**	**+8.0 [+6.6, +9.4]**	**167**
Names in ≥ 7 of 50 samples	17.4	+6.1 [+4.8, +7.4]	111

Team steering at layer 24: +0.7 to +3.0 over random controls on the team's 566 known tune papers. Different papers and labels: not a direct comparison yet.

Stricter label, same picture. Known under the model's own wrong lists (2,481 test papers): default 16.1 per 100; names in ≥ 3 of 20 samples 27.9 (+11.7); CoT +8.5.

Next

1. **Same papers as steering**
   Rerun on split V3's 566 known tune papers
2. **More coverage per token**
   Ask for authors not yet named; first author alone
3. **Better selection**
   Counting plus candidate choice; reachable split at 10, 50 and 200 samples

Generate-then-select needs no model internals; it sets the bar white-box methods should beat at equal compute.